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Result Calculation System — Technical Documentation

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1. ExamResult Data Structure (src/lib/types.ts)

Each student result is stored as an ExamResult object. Below is the full interface with descriptions:

Field	Type	Description
studentId	string	Unique identifier for the student (e.g. s1, s2)
studentName	string	Full name of the student (Latin characters)
class	string	Class ID string (e.g. "class-10", "class-alim-1")
roll	number	Roll number — used as primary key in public result search
subjects[]	array	Array of { name: string, marks: number, grade: string }
totalMarks	number	Sum of marks across all subjects
gpa	number	Overall GPA on a 5.00 scale
grade	string	Overall letter grade (A+, A, A-, B, C, D, F)
status	string	'pass' or 'fail' — determined by GPA and subject grades
examName	string	Name of the exam (e.g. "1st Semester Exam 2024")
year	string	Academic year string (e.g. "2024")

2. Grading Scales — BMEB System (src/lib/data.ts)

The system uses TWO separate grade scales, selected by getGradeScale(classId):

2A. Dakhil / Alim Scale (class-6 to class-10, Alim) — GPA out of 5.00

Marks Range	Grade	GPA	Label
80 – 100	A+	5.00	Excellent (Oti Uttam)
70 – 79	A	4.00	Very Good (Uttam)
60 – 69	A-	3.50	Good (Bhalo)
50 – 59	B	3.00	Good (Bhalo)
40 – 49	C	2.00	Satisfactory (Santoshjanak)
33 – 39	D	1.00	Below Average (Nimmoman)
0 – 32	F	0.00	Fail (Akritakarja)

2B. Ebtedayi Scale (class-1 to class-5) — No GPA, pass mark = 40

Marks Range	Grade	Label
80 – 100	A	Excellent (Oti Uttam)
60 – 79	B	Very Good (Uttam)
40 – 59	C	Satisfactory (Santoshjanak)
0 – 39	D	Needs Support (Sahayata Proyojon)

3. Grade Calculation Logic (src/app/teacher/marks/page.tsx)

When a teacher enters marks, the getGrade() function converts the marks to a letter grade:

```
const getGrade = (m: number): string => {
  if (m >= 80) return "A+"; // Excellent
  if (m >= 70) return "A";  // Very Good
  if (m >= 60) return "A-"; // Good
  if (m >= 50) return "B";  // Satisfactory
  if (m >= 40) return "C";  // Pass
  if (m >= 33) return "D";  // Below Average (pass at Dakhil level)
  return "F";               // Fail
};
```

This function is called once per subject as the teacher saves marks. The grade is stored alongside marks.

4. GPA Calculation & Pass/Fail Rules

4A. GPA per Subject

Grade	GPA Points	Notes
A+	5.00	Marks 80–100
A	4.00	Marks 70–79
A-	3.50	Marks 60–69
B	3.00	Marks 50–59
C	2.00	Marks 40–49
D	1.00	Marks 33–39 (Dakhil/Alim only)
F	0.00	Below 33 — FAIL subject

4B. Overall GPA Formula

```
overallGPA = sum(subjectGPA for each subject) / totalSubjects
```

4C. Pass / Fail Rules

Rule	Condition	Result
F in any subject	Any subject has grade F (marks < 33)	FAIL — regardless of overall GPA
Low overall GPA	Overall GPA < 1.00	FAIL
Dakhil pass	GPA >= 1.00 AND no F grade	PASS
Ebtedayi pass	All subjects >= 40 marks (grade C or above)	PASS

5. Result Publish System (localStorage)

Results are NOT published to a database. The admin toggles visibility per exam using the browser's localStorage.

Step	Action	Technical Detail
1	Admin clicks "Publish Result"	adminresults/page.tsx: togglePublish() called
2	Exam name added to list	localStorage.setItem("published_results_v1", JSON.stringify([...examNames]))
3	Notice auto-created	addNotice() called with pass rate stats — visible to all users
4	Student portal checks	student/result/page.tsx: reads localStorage on mount
5	If exam is in list	Result shown — lock screen hidden
6	If exam NOT in list	Lock icon shown: "Result not yet published"
7	Admin withdraws result	Exam name removed from the list in localStorage

6. Public Result Search (src/app/result/page.tsx)

The public result page (/result) allows anyone to search by roll number, class, and exam name.

```
const found = EXAM_RESULTS.find(r =>
  r.roll === parseInt(roll) && r.class === classId && r.examName === exam
);
```

Note: The search also checks localStorage to ensure the result is published before displaying it.

7. Subjects & Marks by Class (src/lib/data.ts)

Class Level	Subjects	Total Marks	Pass Mark per Subject
Ebtedayi (class-1 to 5)	Quran, Bangla, English, Mathematics, Science	5 × 100 = 500	40 (Grade C)
Dakhil (class-6 to 10)	Quran, Bangla, English, Math, Science, Arabic, Aqaid, History	8 × 100 = 800	33 (Grade D)
Alim (class-alim-1, 2)	Arabic, Quran, Hadith, Fiqh, Bangla, English, Logic, History	8 × 100 = 800	33 (Grade D)
Fazil, Kamil	Advanced Islamic studies subjects	Varies	33 (Grade D)

8. PDF Generation Features

PDF Type	Location	Contents	Orientation
Student Marksheet	src/app/student/result/page.tsx	Institution header, student info, subject table with grades, GPA verdict box, signature lines	Portrait (A4)
Admin Result Sheet	src/app/admin/results/page.tsx	All students, all subjects as columns, color-coded grades, pass/fail, stats header	Landscape (A4)
Attendance Report	src/app/admin/attendance/page.tsx	Day-specific: per-person status table. Monthly: summary with % present per person	Portrait (A4)

All PDFs use jsPDF v4 + jspdf-autotable. Dynamic imports (await import("jspdf")) are used to avoid SSR errors in Next.js.

9. Complete Result Flow — End to End

Step	Who	Action	File
1	Teacher	Logs in to teacher portal	src/app/teacher/page.tsx
2	Teacher	Opens "Marks Entry" page and selects class + exam	src/app/teacher/marks/page.tsx
3	Teacher	Enters marks per student per subject (0–100)	src/app/teacher/marks/page.tsx
4	System	getGrade() assigns letter grade per subject automatically	src/app/teacher/marks/page.tsx
5	System	GPA = average of subject GPAs; status = pass/fail checked	src/lib/data.ts (EXAM_RESULTS)
6	Admin	Reviews results on admin results page	src/app/admin/results/page.tsx
7	Admin	Clicks "Publish Result" — localStorage updated, notice sent	src/app/admin/results/page.tsx
8	Student	Opens student portal result page — sees their result	src/app/student/result/page.tsx
9	Student	Downloads PDF marksheet or prints	src/app/student/result/page.tsx
10	Public	Searches result by roll + class on /result page	src/app/result/page.tsx
11	Admin	Can export full class result sheet as PDF	src/app/admin/results/page.tsx

Key Technical Notes

- Results are stored in src/lib/data.ts as a static EXAM_RESULTS array (not in a database).
- Publish state is stored in browser localStorage — it does not persist across different devices/browsers.